CH 1 — Graphs of Scalar Functions

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1.0 Parametric Curves

We have seen functions that are either explicit or implicit
1. $y = f(x)$
2. $x = y^2$
3. $x^2 + y^2 = 1$

We can also describe a curve using parametric equations where

$$x = x(t), y = y(t), t \in [a, b], \forall a, b \in \mathbb{Z}$$

with $(x(t), y(t))$ is a point of the curve and t is the parameter.

- The curve is called **parametric curve**.
- We get a direction in which the curve is traced as well.

Practices:

1. Find the parametric curve: $x = t - 1, y = 2t + 1, t \in [0, 3]$

   $x + 1 = t \Longrightarrow y = 2(x + 1) + 1 = 2x + 3, x \in [-1, 2]$

2. Find the parametric curve: $x = \cos(t), y = \sin(t), t \in [0, 2\pi]$

   $x^2(t) + y^2(t) = \cos^2 t + \sin^2 t = 1, t \in [0, 2\pi]$

3. Find the parametric curve: $x = \cos(2t), y = \sin(2t), t \in [0, 2\pi]$

   $x^2 + y^2 = \cos^2(2t) + \sin^2(2t) = 1, t \in [0, 2\pi]$

   Same curve as Q2 but has two different parametrizations. Twice as the speed as the Q2.

1.1 Vector-Valued Functions

> 📘 **Definition — Vector-Valued Funciton**
>
> It is a funciton whose domain is a subset of the $\mathbb{R}$ and whose output is a vector

Note: a vector-valued funciton is a convenient way to collect scalar functions

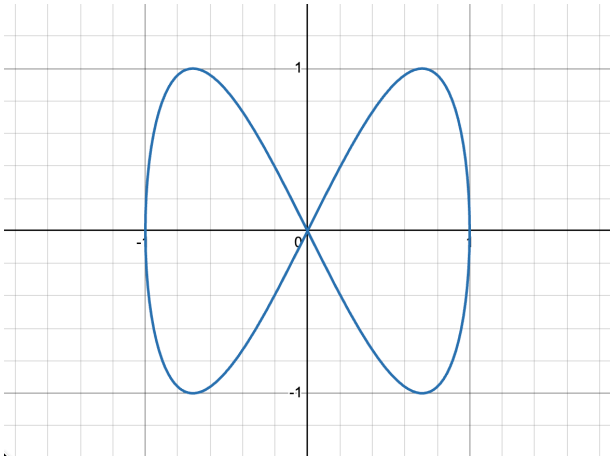
Examples:

1. In $\mathbb{R}^2, \vec{x}(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$

2. In $\mathbb{R}^3, \vec{x}(t) = \begin{bmatrix} x(t) \\ y(t) \\ z(t) \end{bmatrix}$

3. Let $\vec{x}(t) = (\cos t, \sin 2t)$ represent the position of a particle at time t. Sketch $\vec{x}(t), 0 \leq t \leq 2\pi$

   We have $x = \cos t, y = \sin 2t, 0 \leq t \leq 2\pi$



Definition — Velocity

Velocity = $\vec{v}(t) = \frac{\mathrm{d}x(t)}{\mathrm{d}t}$

Examples:

1. $\vec{x}(t) = (\cos t, \sin 2t)$

   velocity $= \frac{\mathrm{d}\vec{x}(t)}{\mathrm{d}t} = (-\sin t, 2\cos 2t)$

   At $t = 0, \vec{v}(0) = (0, 2)$ At $t = \frac{\pi}{2}, \vec{v}(\frac{\pi}{2}) = (-1, 0)$

2. Let $\vec{x}_1(t) = (\cos t, \sin t)$ and $\vec{x}_2(t) = (\cos 3t, \sin 3t)$ be the paths of two particles, find the velocity of each particle when $t = 0$

   $\frac{\mathrm{d}\vec{x_1}(t)}{\mathrm{d}t} = (-\sin t, \cos t)$, 1 revolution of unit circle radius 1 unit

   $\frac{\mathrm{d}\vec{x_2}(t)}{\mathrm{d}t} = (-3\sin 3t, 3\cos 3t)$, 3 revolutions of unit circle radius 1 unit

   $\vec{x_1}'(0) = (0, 1), \vec{x_2}'(0) = (0, 3)$

Definition — Tangent vector

Given the curve defined by

$$\vec{x}(t) = (x(t), y(t))$$

its **tangent vector** at $t = t_0$ is

$$\vec{t} = \left.\frac{\mathrm{d}\vec{x}}{\mathrm{d}t}\right|_{t=t_0} = (x'(t_0), y'(t_0))$$

whenever $\frac{\mathrm{d}\vec{x}}{\mathrm{d}t}, t = t_0 \neq \vec{0}$ exists.

Note: $\vec{t}$ lies on the tangent line and points in the direction of travel

Example:

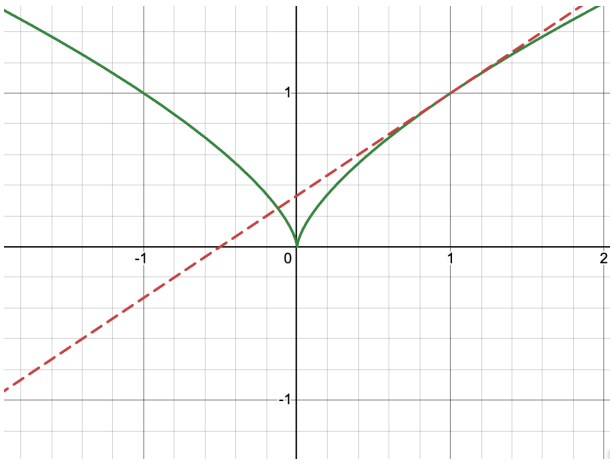
Consider the curve $\vec{x}(t) = (t^3, t^2)$ and its tangent vector.

Notice that if $x = t^3 \longrightarrow t = x^{\frac{1}{3}} \longrightarrow y = x^{\frac{2}{3}}$

$\frac{\mathrm{d}\vec{x}(t)}{\mathrm{d}t} = (3t^2, 2t)$

$t = 1, \vec{x}(1) = (1, 1), \vec{v}(1) = (3, 2)$

$t = 0, \vec{x}(1) = (0, 0), \vec{v}(1) = (0, 0) = \vec{0}$ We cannot define the tangent vector as the function at the point is not differentiable



1.2 Scalar Functions

Recall:

$f : A \to B$ is a function

1. Each element $a \in A$ is a unique element $f(a) \in B$ called the **image** of a under f
2. The set A is called the **domain** of f and is denoted by $D(f)$
3. The set B is called the **codomain** of f.
4. The subset of B consisting of all $f(a)$ is called the **range** of f, denoted by $R(f)$
5. Single variable scalar functions are $f : \mathbb{R} \to \mathbb{R}$

> 📘 **Definition — Scalar Function**
> A **scalar function** $f(x_1, ..., x_n)$ of n-variables is a function whose domain is a subset of $\mathbb{R}^n$ and whose range is a subset of $\mathbb{R}$

Examples:

1. $f(x, y) = x^2 + y^2 \to f(1, 1) = 1^2 + 1^2 = 2$
2. $f(x, y) = \sin(x + y) \to f(0, \frac{\pi}{2}) = \sin(\frac{\pi}{2}) = 1$
3. $f(x, y, z) = e^{x+y+z} \to f(1, 2, 1) = e^{1+2+1} = e^4$

Notation: We can write $\vec{x} = (x_1, x_2, ..., x_n)$ so that $f(\vec{x}) = f(x_1, x_2, ..., x_n)$

Our independent variables don't have to be x, y, z, etc... Multivariable functions can be used to model many different physical situations

Example: write down a multivariable funciton which calculates the volume of a cylinder.

$V = \pi r^2 h \longrightarrow V(r, h) = \pi r^2 h$

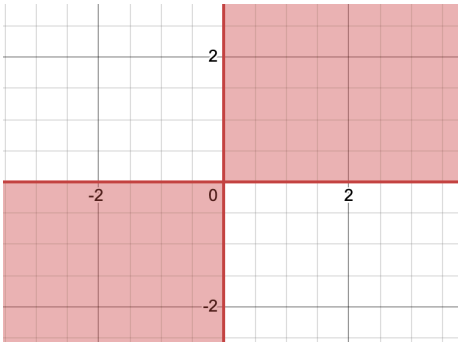
> 📘 **Definition — Domain and Range**
>
> **Domain** is the largest subset of $\mathbb{R}^n$ for which the funciton is defined
>
> **Range** is a set of all possible values which are subset of $\mathbb{R}$

Examples:

1. Find the domain and range of $f(x, y) = \sqrt{xy}$

   We need $xy \geq 0 \implies \begin{cases} x \geq 0 \text{ and } y \geq 0 \\ x \leq 0 \text{ and } y \leq 0 \end{cases}$

   $\text{dom}(f) = \{(x, y) \in \mathbb{R}^2 : x \geq 0, y \geq 0\} \cup \{(x, y) \in \mathbb{R}^2 : x \leq 0, y \leq 0\}$

   We know that $\sqrt{xy} \geq 0, \ \forall x, y \in \text{dom}(f)$

   Let $c \in \mathbb{R}, c \geq 0$, be an arbitrary non-negative real number.

   Notice that $f(c^2, 1) = \sqrt{c^2} = |c| = c$ since $c \geq 0$

   For any $c \in \mathbb{R}, c \geq 0, \exists$ inputs in the domain of f s.t. c is a valid output $\rightarrow c \in \text{ran}(f) : \mathbb{R}_{\geq 0}$

2. Find the domain and range of $g(x, y) = \frac{x^2 - y^2}{|x| + |y|}$

   We need $|x| + |y| \neq 0$ which is true $\forall x, y \in \mathbb{R}$ both note zero since $|x| > 0, |y| > 0$

   $\text{dom}(g) = \{(x, y) : x, y \in \mathbb{R} \setminus \{(0, 0)\}\}$

   Consider case $x > y$: $x^2 - y^2 > 0, |x| + |y| > 0 \implies g(x, y) > 0$

   Consider case $x = y$: $x^2 - y^2 = 0, |x| + |y| = 2x \implies g(x, y) = 0$

   Consider case $x < y$: $x^2 - y^2 < 0, |x| + |y| > 0 \implies g(x, y) < 0$

   or alternatively

   Let $c \in \mathbb{R}, c > 0, g(c, 0) = \frac{c^2 - 0}{|c| + 0} = |c| = c, c > 0$

   Let $d \in \mathbb{R}, d < 0, g(0, d) = \frac{0 - d^2}{0 + |d|} = |d| = d, d < 0$

   Let $a, b \in \mathbb{R}, a = b, g(a, b) = \frac{a^2 - b^2}{|a| + |b|} = \frac{a^2 - a^2}{|a| + |a|} = \frac{0}{2a} = 0$

   Thus $\text{ran}(g) = \mathbb{R}$

1.3 Geometric Interpretation of $z = f(x, y)$

Question: How we can graph $z = f(x, y)$

ANS: We graph $(a, b, f(a, b))$. We think of $f(a, b)$ as height above the xy-plane

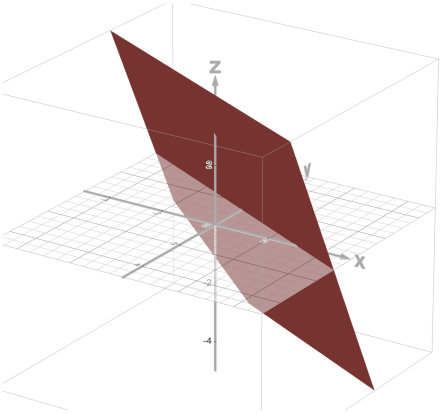
> 📘 **Definition — Level Curves**
>
> The **level curves** of a function $f(x, y)$ are the curves
>
> $$f(x, y) = k$$
>
> where k is a constant in the range of f
>
> - **Alternate definition**: curves along which the height of the graph $z = f(x, y)$ is constant
> - A collection of **level curves** are called a **contour map**
> - There are **exceptional level curves** because it looks different from the other level curves.

> 📘 **Definition — Cross Sections**
>
> A **cross section** of a surface $z = f(x, y)$ is the intersection of $z = f(x, y)$ with a plane

Examples:

1. Sketch $x + 2y + z = 3$

   We can rewrite this as $z = 3 - x - 2y = f(x, y)$

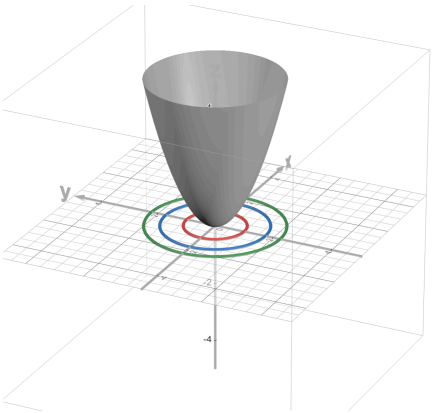
   

   Sketch and write down an equation which is the level curves for the above and sketch these level curves for a different k

   Level curve: $3 - x - 2y = k, k \in \mathbb{R} \implies y = -\frac{1}{2}x + \frac{1}{2}(3 - k)$

2. Sketch $f(x, y) = x^2 + y^2$

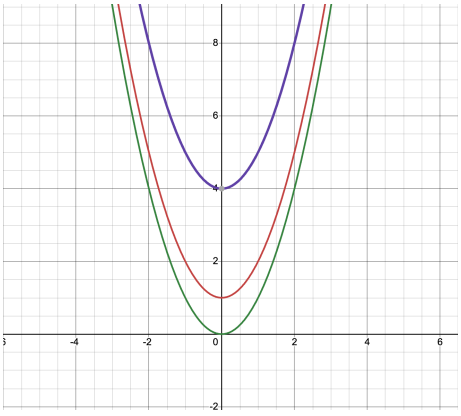
   Level curves: $x^2 + y^2 = k, k \geq 0$ (red circle below)



This is a paraboloid(grey “cone-like” shape)

3. $z = x^2 + y^2$

   $x = c \implies z = c^2 + y^2$

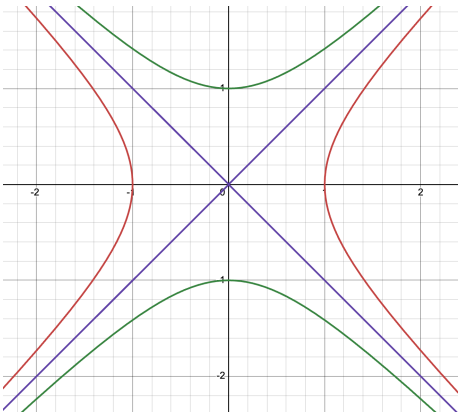


   $y = d \implies z = x^2 + d^2$ (Similar like above)

4. $z = x^2 - y^2$

   $\text{dom}(f) = \mathbb{R}, \text{ran}(f) = \mathbb{R}$

   Level curves: $x^2 - y^2 = k, k \in \mathbb{R}$ (hyperbolas)



   Red lines $k > 0$
   Green lines $k < 0$
   Blue lines $k = 0$

📘 **Definition — Levels**

1. A **level surface** of a scalar function $f(x, y, z)$ is defined by

$$f(x, y, z) = k, \quad k \in R(f)$$

2. A **level set** a scalar function $f(x), x \in \mathbb{R}^n$ is defined by

$$f(x) = k, \quad k \in R(f)$$